Department of Computer Science and Engineering

Nandha Engineering College (Autonomous) – Erode.

Project Review

A Comparative Transformer-Based Framework for Automated Classification of Pterygium and Conjunctivitis

Presented By,

PREETHISHRI D (24CSF05)

II-ME-CSE

Under the Guidance of,

.

Dr. SIVARAJ R
Professor
Department of Computer Science and Engineering,
Nandha Engineering College (Autonomous), Erode- 638052.

ABSTRACT

This study proposes a transformer-based deep learning framework for the automated classification of anterior segment ocular diseases, specifically pterygium and conjunctivitis, using heterogeneous clinical image datasets. Four advanced vision transformer architectures such SegFormer, Vision Transformer (ViT), MaxViT, and MobileViT—are systematically evaluated using transfer learning to enhance feature generalization. Unlike conventional convolutional models, transformer-based approaches effectively capture long-range spatial dependencies and contextual relationships within ocular images. Among the evaluated models, SegFormer achieves the highest classification accuracy of 96%, outperforming ViT (95%), MaxViT (92%), and MobileViT (94%). Cross-dataset validation further demonstrates strong generalization, with SegFormer attaining 90% accuracy for pterygium and 95% for conjunctivitis. The study highlights the effectiveness of lightweight and scalable transformer architectures for reliable and clinically applicable ocular disease diagnosis, contributing to the development of AI-driven clinical decision support systems.

INTRODUCTION

- Deep learning has revolutionized medical image analysis, enabling automated and accurate disease detection across various clinical domains, including ophthalmology.

- Anterior segment diseases such as pterygium and conjunctivitis exhibit subtle visual differences, making manual diagnosis subjective and dependent on clinical expertise.

- Conventional convolutional neural networks struggle to capture long-range spatial dependencies and global contextual information in complex ocular images.

- Vision transformers leverage self-attention mechanisms to model global relationships, offering improved performance in medical image classification and multi-disease detection tasks.

OBJECTIVES

- To design an automated system for classifying pterygium and conjunctivitis using advanced vision transformer architectures.

- To evaluate and compare the performance of SegFormer, ViT, MaxViT, and MobileViT under a unified experimental setup.

- To validate model performance across multiple heterogeneous datasets for improved generalization and reduced bias in clinical deployment.

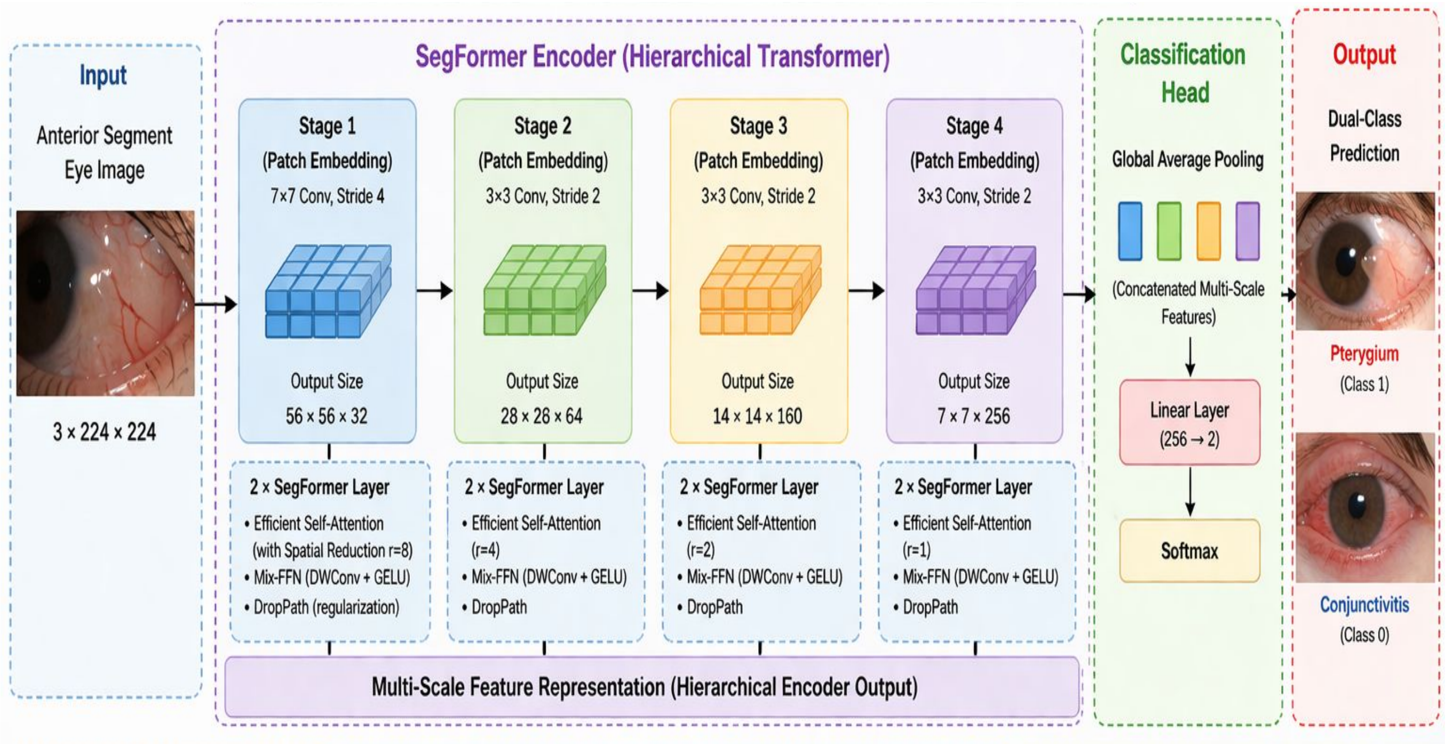
LITERATURE SURVEY

Paper Tilte	Conference Applied	Overview of Existing Work	Shortcomings	Justification for Proposed Approach
Deep learning for diagnosing and grading pterygium: A systematic review and meta-analysis	Computers in Biology and Medicine (2025)	Comprehensive review of deep learning models for pterygium detection and grading	Focuses on review only; lacks implementation and comparative analysis	Motivates need for practical model comparison using advanced architectures
Semantic segmentation-based automatic pterygium system	Frontiers in Medicine (2025)	Uses segmentation techniques for pterygium grading	Limited to single disease; lacks classification generalization	Supports need for multi-disease classification framework
AI assisted pterygium diagnosis	International Journal of Ophthalmology (2023)	Discusses AI applications in pterygium diagnosis	Lacks experimental benchmarking and multi-model comparison	Justifies comparative evaluation of multiple transformer models
Conjunctivitis detection using CNN [9]	Applied Data Science (2024)	CNN-based conjunctivitis detection	Limited performance due to CNN constraints	Justifies use of transformer models for better feature extraction
Deep learning for conjunctivitis [10]	ICOECA Conference (2025)	Applies deep learning for conjunctivitis classification	Single-disease focus; lacks comparative study	Motivates dual-class classification system

PROPOSED METHODOLOGY

- Four transformer-based models such as SegFormer, Vision Transformer (ViT), MaxViT, and MobileViT—were fine-tuned for binary classification of anterior segment ocular images into pterygium and conjunctivitis using transfer learning and multi-dataset training.
- **SegFormer** – Utilized a hierarchical transformer encoder with efficient self-attention and multi-scale feature extraction; achieved 96% accuracy, the highest among all models, with strong generalization across datasets.
- **Vision Transformer (ViT)** – Pure transformer model leveraging global self-attention on image patches; achieved 95% accuracy, effectively capturing global context but limited in fine-grained local feature representation.
- **MaxViT** – Hybrid convolution-transformer architecture with multi-axis attention (local and global); achieved 92% accuracy, showing good contextual learning but higher complexity affecting performance on smaller datasets.
- **MobileViT** – Lightweight hybrid model combining CNN and transformer blocks for efficient feature learning; achieved 94% accuracy, offering a balance between performance and computational efficiency suitable for real-time applications.

PROPOSED ARCHITECTURE



IMPLEMENTATION

Modules and their integration

Dataset Collection & Integration

- Multiple heterogeneous clinical image datasets of anterior segment ocular diseases are collected.
- Images include two classes: pterygium and conjunctivitis.
- Data diversity ensures real-world variability and robustness.

Data Preprocessing

- Image resizing to 224 × 224 resolution for model compatibility.
- Normalization to standardize pixel intensity values.
- Data augmentation techniques such as rotation, flipping, and intensity variation to improve generalization.

Model Selection (Transformer Architectures)

- Four advanced models are selected:
  - SegFormer
  - Vision Transformer (ViT)
  - MaxViT
  - MobileViT

Transfer Learning & Fine-Tuning

- Backbone layers are initialized with pretrained weights.
- Final classification layer is modified for binary classification.
- Fine-tuning improves domain-specific learning.

Model Training

- Optimized using Adam optimizer and cross-entropy loss.
- Training performed over 25 epochs with mini-batch learning.

Performance Evaluation

- Metrics used: Accuracy, Precision, Recall, and F1-score.
- Comparative analysis across all four models.

Cross-Dataset Validation

- Models tested on independent datasets to evaluate generalization capability.
- Ensures robustness against dataset variability.

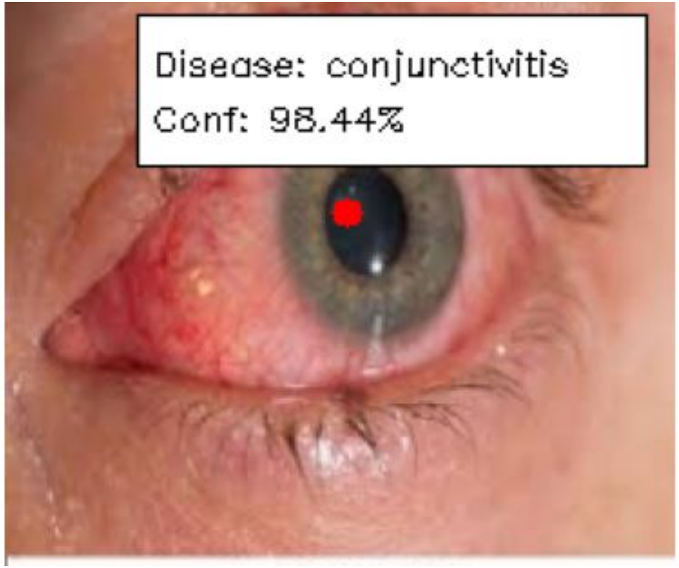
Interpretability (Grad-CAM Analysis)

- Visual explanations generated to highlight important regions in images.
- Enhances model transparency and clinical trust.

RESULTS AND ANALYSIS

The experimental evaluation of the proposed transformer-based framework demonstrates that all four models achieve strong performance in classifying anterior segment ocular diseases, with noticeable variations in accuracy and generalization capability. Among the evaluated architectures, SegFormer outperforms the others by achieving the highest overall classification accuracy of 96%, indicating its superior ability to capture both local and global contextual features. Cross-dataset validation highlights the robustness of SegFormer, achieving 90% accuracy for pterygium and 95% for conjunctivitis, thereby demonstrating strong generalization across heterogeneous clinical data. The results confirm that transformer-based models, particularly hierarchical architectures like SegFormer, are highly effective for ocular disease classification and offer significant advantages over traditional approaches in terms of accuracy, scalability, and real-world applicability.

OUTPUT SAMPLES



APPLICATIONS / USE CASES

- ➢ The proposed automated ocular disease classification system has significant applications in clinical diagnostics, particularly for the early detection of pterygium and conjunctivitis in diverse healthcare settings.
- ➢ It can be integrated into telemedicine platforms to facilitate remote screening and assist healthcare providers in prioritizing patients based on anterior segment images.
- ➢ In hospitals and eye care centers, the system can support ophthalmologists by offering a reliable second opinion, thereby improving diagnostic accuracy, speed, and consistency.
- ➢ The framework is well-suited for deployment in mobile health (mHealth) applications when combined with portable imaging devices, enabling point-of-care diagnosis in rural and underserved regions.
- ➢ The proposed approach demonstrates strong potential for scalable implementation in AI-driven clinical decision support systems, enhancing accessibility and efficiency in ophthalmology.

CONCLUSION

The present study successfully demonstrates the effectiveness of transformer-based deep learning models for the automated classification of anterior segment ocular diseases, specifically pterygium and conjunctivitis. Among the evaluated architectures, SegFormer achieves the highest performance with 96% accuracy, highlighting its superior capability in capturing both local and global feature representations. Comparative analysis confirms that transformer models outperform traditional approaches by effectively modeling long-range dependencies and contextual information. The incorporation of multi-dataset validation further ensures robustness and generalization across diverse clinical conditions. Overall, the proposed framework offers a reliable, scalable, and clinically applicable solution, paving the way for the development of AI-driven decision support systems for early and accurate ocular disease diagnosis.

REFERENCES

[1] E. W. W. Tiong, C. Y. S. Soon, Z. Z. Ong, S.-H. Liu, R. Qureshi, S. Rauz, and D. S. J. Ting, “Deep learning for diagnosing and grading pterygium: A systematic review and meta-analysis,” *Computers in Biology and Medicine*, vol. 196, pt. A, p. 110743, Sep. 2025, doi: 10.1016/j.compbiomed.2025.110743.

[2] Q. Ji, W. Liu, Q. Ma, L. Qu, L. Zhang, and H. He, “A semantic segmentation-based automatic pterygium assessment and grading system,” *Frontiers in Medicine*, vol. 12, 2025, doi: 10.3389/fmed.2025.1507226.

[3] B. Chen, M.-N. Wu, S. Zhu, and B. Zheng, “Artificial intelligence assisted pterygium diagnosis: Current status and perspectives,” *International Journal of Ophthalmology*, vol. 16, no. 9, pp. 1386–1394, Sep. 2023, doi: 10.18240/ijo.2023.09.04.

[4] Z. Xu, J. Xu, C. Shi, W. Xu, X. Jin, W. Han, K. Jin, A. Grzybowski, and K. Yao, “Artificial intelligence for anterior segment diseases: A review of potential developments and clinical applications,” *Advances in Therapy*, vol. 12, no. 3, pp. 1439–1455, Mar. 2023, doi: 10.1007/s40123-023-00690-4.

[5] D. S. Kermany *et al.*, “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, Feb. 2018, doi: 10.1016/j.cell.2018.02.010.

[6] K. Jin and J. Ye, “Artificial intelligence and deep learning in ophthalmology: Current status and future perspectives,” *Advances in Ophthalmology Practice and Research*, vol. 2, no. 3, p. 100078, Nov.–Dec. 2022, doi: 10.1016/j.aopr.2022.100078.

[7] J. Son, J. Y. Shin, H. D. Kim, K.-H. Jung, K. H. Park, and S. J. Park, “Development and validation of deep learning models for screening multiple abnormal findings in retinal fundus images,” *Ophthalmology*, vol. 127, no. 1, pp. 85–94, Jan. 2020, doi: 10.1016/j.ophtha.2019.05.029.

[8] R. Sakata *et al.*, “Deep learning-based diagnostic model for ocular surface neoplastic diseases,” *American Journal of Ophthalmology*, 2026, doi: 10.1016/j.ajo.2026.02.033.

THANK YOU